

PYTHON FUNDAMENTALS

INFO1 Mission 2 (S3)

Jean-Luc Bibaud



Type and Value

```
>>> type(5)
<class 'int'>
>>> type(3.14)
<class 'float'>
>>> type("Ada")
<class 'str'>
>>> type(True)
<class 'bool'>
```

`type(value)` → its type

5

→ int

3.14

→ float

"Ada"

→ str

True

→ bool

Concatenation vs Addition

Addition

```
1 + 1
```

= **2** numbers are added

Concatenation

```
"hi " + "world"
```

= **"hi world"** strings are joined

Which branch runs?

```
if x < y:  
    print("STATEMENTS_A")  
elif x > y:  
    print("STATEMENTS_B")  
else:  
    print("STATEMENTS_C")
```

In this example, what happens if we have

```
x = 3  
y = 4
```

Explicit vs Implicit type conversion

Explicit

```
A = str(55.55)
```

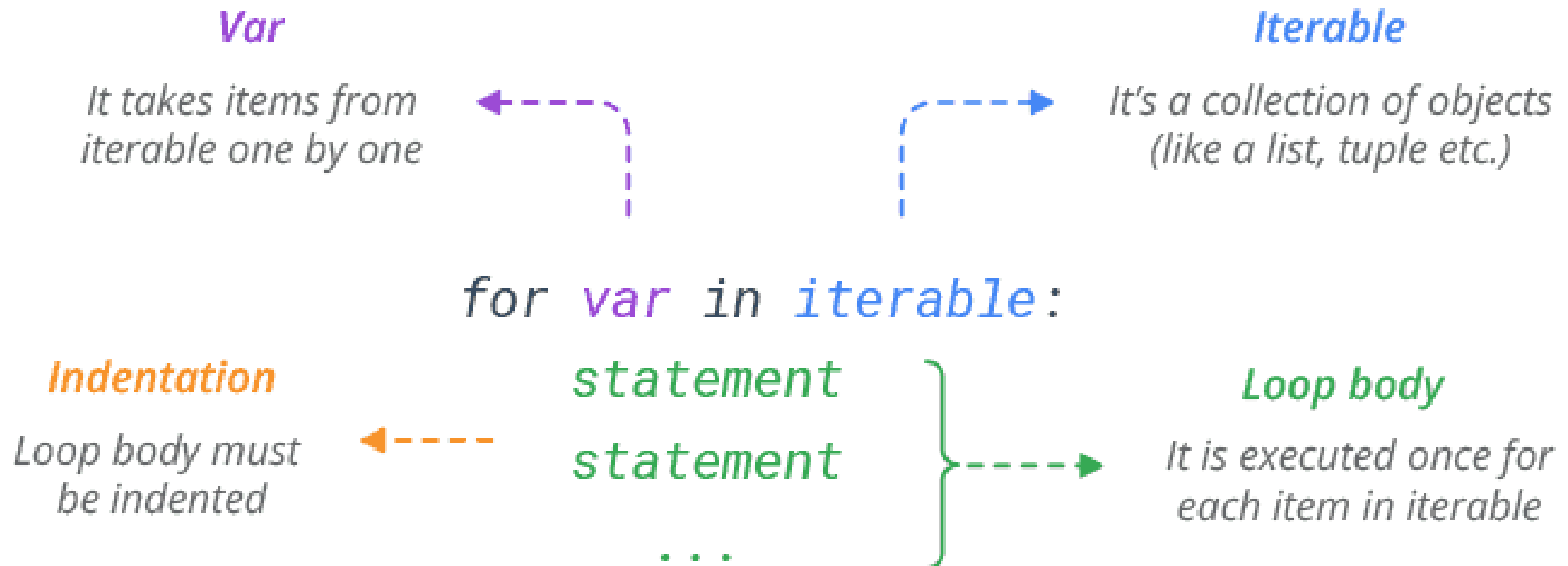
you ask for the conversion

Implicit

```
A = 2  
A = A + 0.14
```

Python converts for you (int → float)

For loops



Range for making numbers Iterable

```
range(start, stop, step)
```

range(stop) is the same as **range(0, stop, 1)**

Break to get out of a loop

```
for i in range(5):  
    print(i)  
    if i == 3:  
        break
```

A

0
1
2

B

0
1
2
3

C

0
1
2
3
4

Continue to skip an iteration

```
for i in range(5):  
    print(i)  
    if i == 3:  
        continue
```

A

0
1
2
3
4

B

0
1
2
3

C

0
1
2